6.2 Radicals and Rational Exponents

Name _____ Period ____ Date _____

Algebra I 6.2 · Student sheet

I can…

- find nth roots.
- rewrite expressions between radical form and rational exponent form.
- evaluate expressions with rational exponents.

Vocabulary · fill in the missing word

nth root — if $b^n = a$, then b is an nth root of a, _____ $\sqrt[n]{a}$

Index — _____

Rational exponent — $a^{\left(\frac{1}{n}\right)}$ = _____ = $\left(\sqrt[n]{a}\right)^m$

Part 1 · Exponent detective

$5^{(1/2)} \cdot 5^{(1/2)} = 5^1 = 5$. So $5^{(1/2)}$ is the number that times itself makes 5. That number is _____

$8^{(1/3)} \cdot 8^{(1/3)} \cdot 8^{(1/3)} = 8^1 = 8$. So $8^{(1/3)}$ = _____

Expression	9^(1/2)	16^(1/2)	8^(1/3)	27^(1/3)	16^(1/4)
Value					

Rule: a^(1/n) means _____

Part 2 · Notes: fill in the blanks

$\sqrt[n]{a}$ is a number whose _____ power is a. The n is called the _____.

If n is odd, a has _____ real nth root. If n is even and a > 0, a has _____ real nth roots.

$a^{(1/n)}$ = _____ $a^{(m/n)}$ = _____

In $a^{(m/n)}$, the denominator tells the _____ and the numerator tells the _____.

Part 3 · Examples (work them with the class)

Example 1

Rewrite each expression in radical form.

a. $10^{\left(\frac{1}{2}\right)}$ b. $11^{\left(\frac{1}{5}\right)}$

Example 2

Find the indicated real nth root(s) of a.

a. n = 3, a = −27 b. n = 4, a = 16

Example 3

Evaluate each expression.

a. $\sqrt[3]{(-64)}$ b. $-\sqrt[4]{16}$

Example 4

Evaluate each expression.

a. $16^{(\frac{3}{4})}$ b. $27^{(\frac{2}{3})}$ c. $32^{(-\frac{2}{5})}$

Example 5

A cube has a volume of 1331 cubic feet. Find the length of each side.

Part 4 · Partner practice

Partner 1

Evaluate $64^{(\frac{2}{3})}$.

Partner 2

Evaluate $81^{(-\frac{1}{4})}$.